

Coordination bonds

Learning outcomes

By the end of this section you should be able to:

- Identify coordination bonds in compounds.
- Describe the formation of coordination bonds and their importance.

Imagine that you forgot your lunch going to school. Luckily, your friend has packed extra food that day and so you both decide to share their lunch. Both of you eat the shared lunch and both of you feel full after. If the aim of having lunch is to feel full, does it matter where the food came from?

Coordination bonds are based on a similar idea. These bonds form in scenarios where one species has a pair of electrons it can share when bonding and another species is unable to donate any electrons to a shared pair. If the aim of a covalent bond is to achieve stability through sharing electrons, does the origin of the electrons matter?

Coordination bonding

Remember from section S2.2.1 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43106/>) that in a covalent bond, each atom being bonded donates one electron each to the bond. A coordination bond is a type of covalent bond formed between two atoms where both electrons in the pair being shared have been provided by just one atom (**Figure 1**). This does not affect the properties of the covalent bond formed; coordination bonds have the same properties as a regular covalent bond. The only difference is in the origin of the electrons.

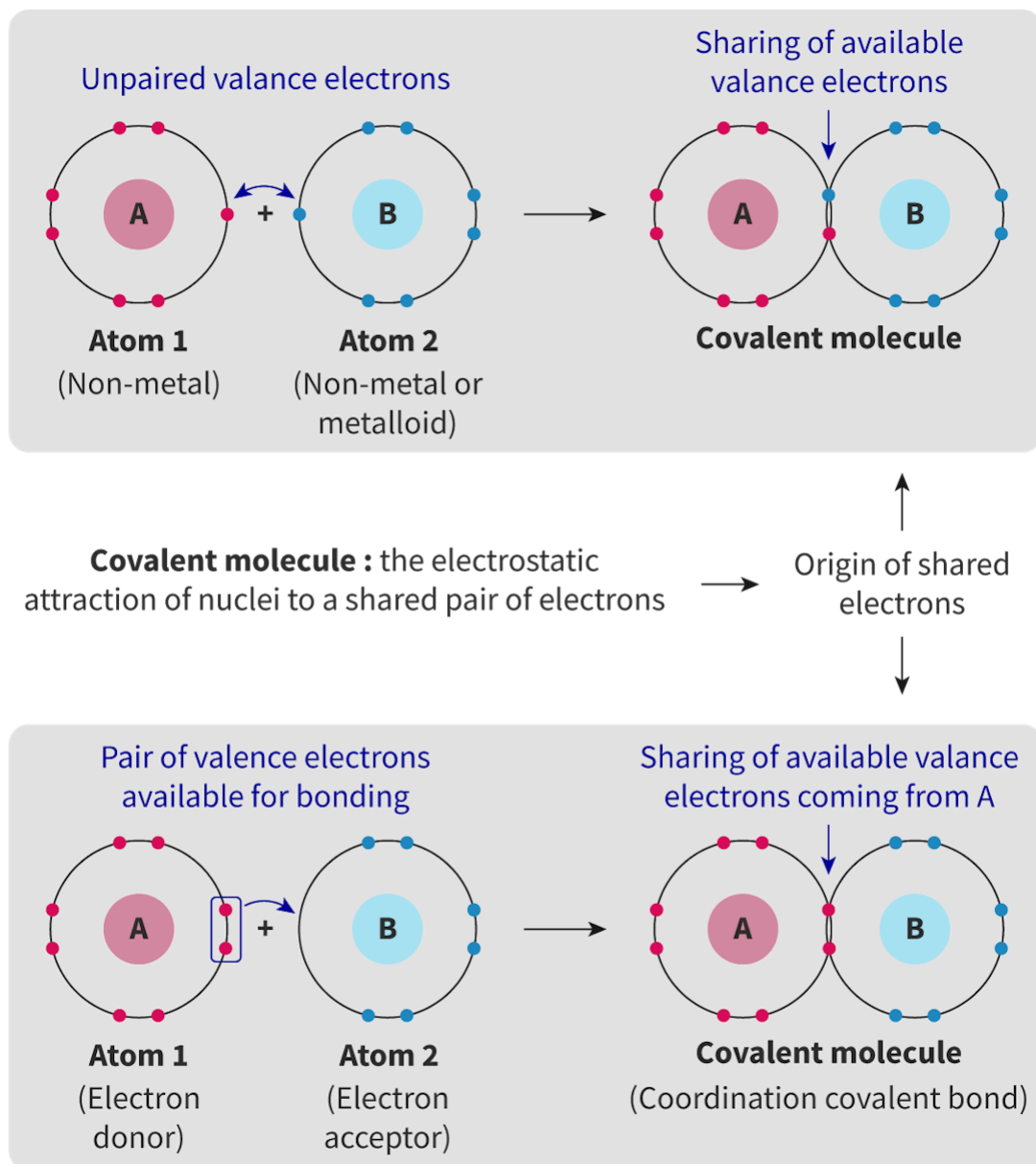


Figure 1. Coordination bonds are a specific type of covalent bond.

 More information for figure 1

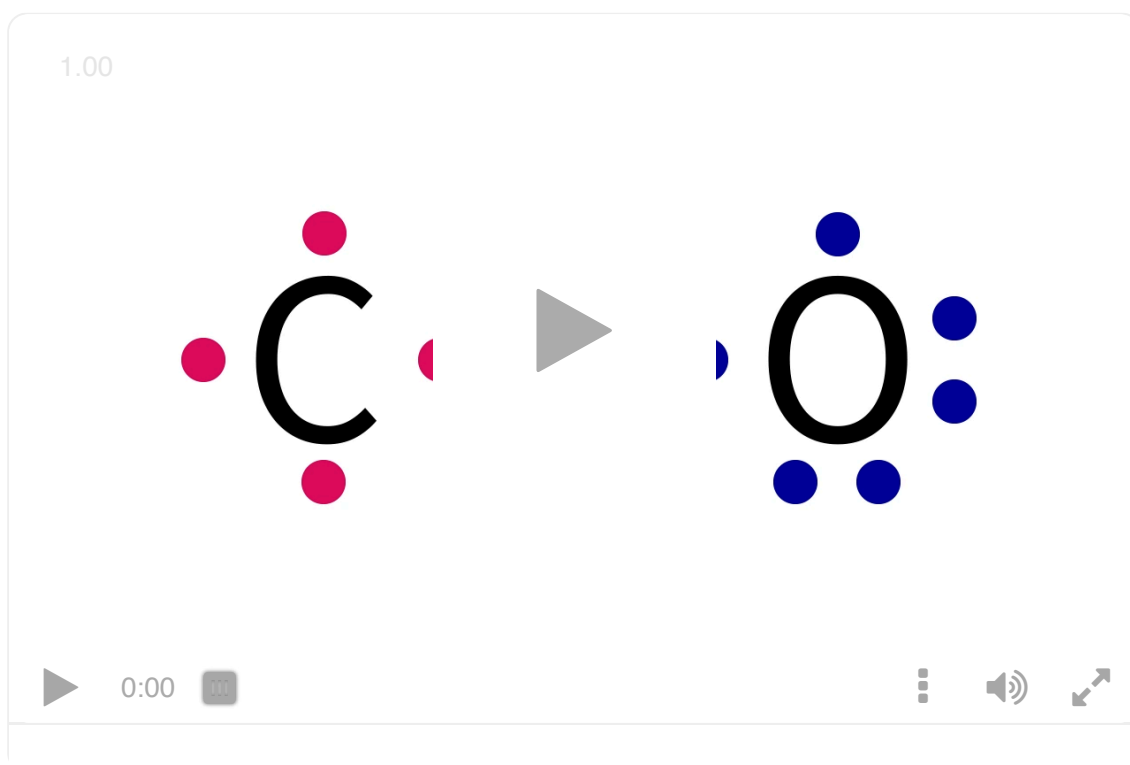
The image is a diagram illustrating covalent and coordination covalent bonds between two atoms, labeled as Atom 1 (A) and Atom 2 (B). On the left side, it shows the formation of a covalent bond. Atom 1, a non-metal, and Atom 2, a non-metal or metalloid, each have unpaired valence electrons. The diagram indicates that these electrons are shared, forming a covalent molecule. The text explains that a covalent molecule is the electrostatic attraction of nuclei to a shared pair of electrons.

On the right side, the diagram explains a coordination covalent bond. Here, Atom 1 is labeled as the electron donor with a pair of valence electrons available for bonding. Atom 2 is labeled as the electron acceptor. The available electrons from Atom 1 are shared with Atom 2, forming a coordination covalent bond. The resulting molecule is also referred to as a covalent molecule, with the note that the sharing of valence electrons originates from Atom 1.

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To form a coordination bond one of the species involved in the bond must have non-bonding electrons available for bonding. This atom becomes known as the donor. The other chemical species involved, the acceptor, must have space to accept the electrons (recall the species with incomplete octets that you learned about in [section S2.2.1 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43106/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43106/)). Coordination bonds allow atoms in a molecule to achieve a stable configuration with eight electrons in their outermost shell, which is in accordance with the octet rule.

Carbon monoxide is an example of a molecule that would not form unless a coordination bond was involved (**Interactive 1**). Carbon and oxygen can each contribute two electrons to forming the double bond and although this would allow for oxygen to fulfil the octet rule, as carbon would only have six valence electrons, it would not obey the octet rule. By oxygen donating another non-bonding pair of electrons to form another bond, a triple bond in total, both carbon and oxygen now fulfil the octet rule.



Interactive 1. Coordination Bonds in Carbon Monoxide.

More information for interactive 1

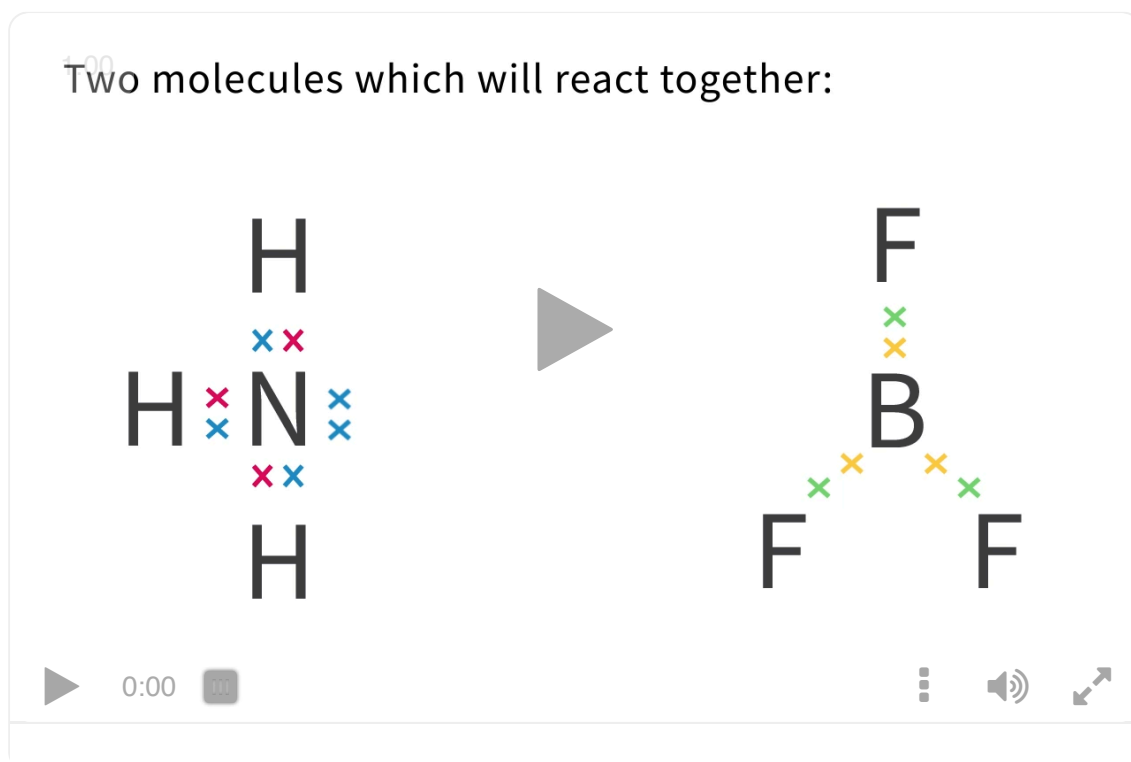
The video begins with the electron configuration of carbon and oxygen. Carbon (C), on the left, has four valence electrons, while oxygen (O), on the right, has six valence electrons. They slowly move towards each other. One electron from each element bond forms a bond. Onscreen text: Single bond forms, both elements not fulfilling octet rule. The electrons reconfigure themselves such that there is a pair of electrons of each element in the center. The onscreen text reads: Double bond forms, oxygen fulfils octet rule but carbon doesn't.

The electrons again reconfigure themselves such that the oxygen donates four electrons and carbon donates two electrons for bond formation. The onscreen text reads as follows: Triple bond forms as oxygen donates a non-bonding pair to make a coordination bond; both atoms now fulfil the octet rule.

A structure displays a triple bond between C and O where two bonds are drawn as lines and the third is represented as an arrow pointing from O to C, representing one of the bonds. The text reads, Representing the coordination bond.

This interactive helps users understand the coordination bond between carbon and oxygen.

Another example is the compound formed in the reaction between ammonia, NH_3 , and boron trifluoride, BF_3 , in **Interactive 2**. Unless ammonia was able to donate a non-bonding pair of electrons to the bond, this compound could not form.



Interactive 2. Coordination bond forming between the nitrogen atom in ammonia and the boron atom in boron trifluoride.

 More information for interactive 2

An interactive video demonstrates the formation of a coordinate bond between an ammonia molecule (NH_3) and a boron trifluoride (BF_3) molecule using Lewis structures.

Ammonia consists of a central nitrogen (N) atom bonded to three hydrogen (H) atoms at the top, left, and bottom. Each bond consists of a pair of shared electrons, one from nitrogen and one from hydrogen. Additionally, nitrogen has a lone pair of electrons.

The electrons are represented as crosses in both molecules.

Boron trifluoride consists of a central boron (B) atom bonded to three fluorine (F) atoms at the top, bottom right, and bottom left. Each bond consists of a pair of shared electrons, one from boron and one from fluorine.

As the video progresses, the two molecules move closer together, and nitrogen donates its lone pair to boron, forming a coordinate bond.

This interactivity helps users understand how nitrogen donates its lone pair to boron, forming a coordinate bond, and completing boron's octet.

Representing coordination bonds

In sections S2.2.1a (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43106/>) and S2.2.2 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/multiple-covalent-bonds-id-45128/>), you used Lewis formulas to represent valence and bonded electrons in molecules. You should also use that same notation to represent coordination bonds, similar to how the previous molecules were represented.

An alternative notation for showing a coordination bond demonstrates the origin of the electrons using an arrow pointing from the donor to the acceptor (**Figure 2**).

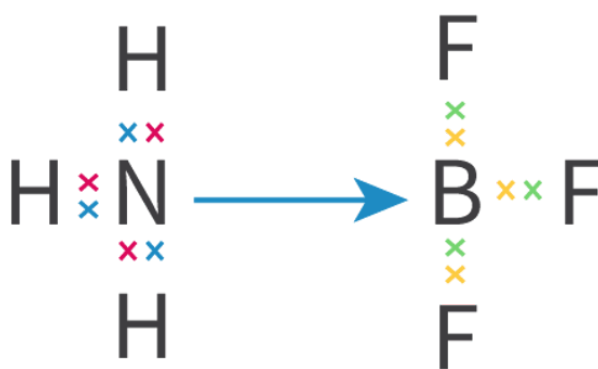
For example, consider the compound formed between ammonia and boron trifluoride (**Interactive 2**), which we looked at previously.



 More information

The image is a Lewis structure diagram showing a complex of NH_3 and BF_3 . In the center, a Nitrogen (N) atom is bonded to three Hydrogen (H) atoms forming NH_3 , with a lone pair of electrons depicted as a pair of blue x's. This Lewis structure shows the NH_3 molecule donates its lone pair to form a dative bond (shown with a blue arrow) with a Boron (B) atom in BF_3 . The Boron atom is connected to three Fluorine (F) atoms. Each Hydrogen and Fluorine is depicted with their bonds represented by lines, and lone pairs on Fluorines are shown as pairs of x's, with different colors for clarity. The structure illustrates the coordinate covalent bond between NH_3 and BF_3 clearly, highlighting the electrons involved in bonding and lone pairs.

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 More information

The image is a chemical diagram illustrating a reaction between ammonia (NH_3) and boron trifluoride (BF_3). On the left side, there is an ammonia molecule represented with a nitrogen atom (N) at the center, bonded to three hydrogen atoms (H). The nitrogen atom has an excess pair of electrons depicted as blue and red crosses. To the right, the molecule boron trifluoride is shown, where a boron atom (B) is at the center bonded to three fluorine atoms (F). There are arrows pointing from the excess electrons in ammonia to the boron atom in boron trifluoride, illustrating the formation of a coordinate covalent bond. The arrow indicates the direction of electron flow from the nitrogen atom to the boron atom.

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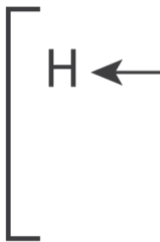
Figure 2. Representing the coordination bond between ammonia and boron trifluoride.

Using the colour of the electrons highlights how the electrons used to form the bond have come from the nitrogen. This can also be represented using an arrow, showing electron movement from donor to acceptor. The arrow should point towards the species that is the recipient of the electron pair, as in **Figure 2**.

Worked example 1

Draw the compound formed from the reaction between a water molecule, H_2O and a hydrogen ion, H^+ .

Solution steps	Calculations
Step 1: Draw the Lewis diagram.	 and 
Step 2: Identify if a coordination bond can be formed.	

Solution steps	Calculations
Final Answer: Represent the final molecule making clear that a coordination bond has been formed using a straight arrow, pointing towards the atom which did not donate the electrons.	

Study skills

In other sources you may encounter other terms for coordination bonds. They can also be called a dative bond, a dipolar bond or a coordinate covalent bond. In your answers, you should always use the term coordination bond. However, be prepared to see these three terms used interchangeably in other sources.

Coordination bonds also exist in compounds and complexes involving transition elements. See [section S3.1.8 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-2-id-43595/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-2-id-43595/) (HL only) for information on these complexes.

Recall the ‘shared lunches’ analogy used at the beginning of this section to introduce the concept of coordination bonds. In this next activity, you will create your own analogy for coordination bonds and display it in a video format.

Activity

- **IB learner profile attribute:** Risk-taker
- **Approaches to learning:** Communication skills — Using digital media for communicating information
- **Time required to complete activity:** 20 minutes
- **Activity type:** Group activity

Creating a short film

Consider the 'shared lunches' analogy used at the beginning of this subtopic. In groups of 2—4, suggest two other analogies that could be used to represent coordination bonds. Share these with the class in the form of a presentation, short video or skit lasting no more than 2 minutes. Use the checkboxes below to ensure that you include all the necessary components.

Option 1: presentation. Make sure to:

- Be clear and follow a logical order.
- Be informative and engaging.
- Have colourful slides which include a mix of both words and images.
- Include two analogies.
- Include an explanation of a coordination bond using correct scientific terminology.
- Discuss at least one example of a coordination bond.

Option 2: video. Make sure to:

- Be clear and follow a logical order.
- Be colourful and engaging.
- Include two analogies.
- Include an explanation of a coordination bond using correct scientific terminology.
- Illustrate at least one example of a coordination bond.

Option 3: skit. Make sure to:

- Be clear and follow a logical order.
- Be informative and engaging.
- Include two analogies.
- Include an explanation of a coordination bond using correct scientific terminology.
- Discuss at least one example of a coordination bond.